

Statistical Mechanics I Exercise 4

Idan Stark

January 27, 2026

1 Landau Theory For The Tri-critical Point

In this section we consider the following model that has a tri-critical point:

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - K \sum_{\square} \sigma_i \sigma_j \sigma_k \sigma_l - h \sigma_i \quad (1)$$

1.1 The phase diagram

To draw the phase diagram, let us notice a few things about the model:

- For $K = 0$, the model is reduced to the Ising model, meaning that at that point we get the well known second order phase transition.
- K is an interaction term, and so will have a similar effect on the critical temperature to that of J . So, $T^*(K)$ has a positive slope.
- Then, of course, the area $T < T^*(K)$ is the ordered phase with multiple solutions, while $T > T^*(K)$ is the unordered phase with a global minimum at $m = 0$.
- Since higher temperature pushes the system towards the center to $m = 0$, the line that separates the 3 solutions area from the five solutions area $K^*(T)$ has positive slope as well: the higher the temperature, the higher K^* needs to be to pass create the two additional solutions.
- Finally, the tricritical point is at the meeting of these two lines.

The phase diagram in the $K - T$ space can be seen in figure 1.

1.2 The Landau free energy

The Landau free energy can be seen in figure 2 at different parts of the phase space. As can be seen, at some parts of the phase space the free energy has five extrema. This is possible only for a polynomial with degree six or above. Since for $h = 0$ the model retains the $m \rightarrow -m$ symmetry, the polynomial has only even powers except for the linear term with prefactor h , leading to the final form of the free energy:

$$\Phi = -hm + am^2 + bm^4 + dm^6 \quad (2)$$

1.3 Critical exponents

Let us calculate the critical exponents α and β near the tricritical point. At that point, For these two exponents, we set $h = 0$ and derive to find the minimising magnetization:

$$\frac{\partial \Phi}{\partial m} = 2am + 4bm^3 + 6dm^5 = 0 \quad (3)$$

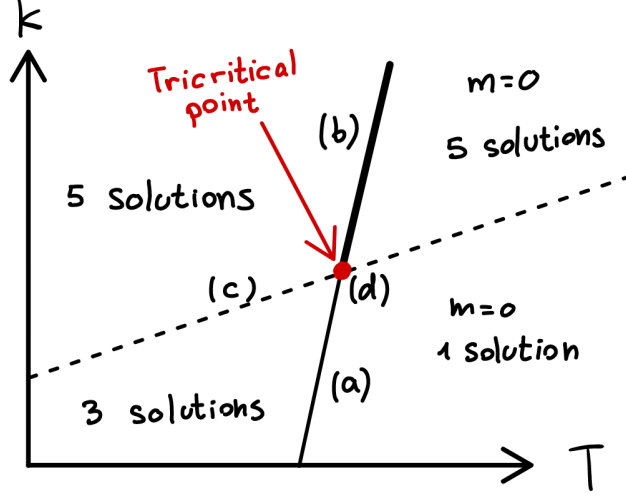


Figure 1: The phase diagram of the model in the $K - T$ space. (a) The second order phase transition line, (b) the first order phase transition line, (c) the line separating the 3-solutions and 5-solutions areas (d) the tricritical point

$$m = 0, m^2 = \frac{-b \pm \sqrt{b^2 - 3ad}}{3d} \quad (4)$$

At the tricritical point, all solutions converge, which requires $a = b = 0$. So, going along the line $d = \text{const}$ we can expand a and b as a function of $t = \frac{T-T_C}{T_C}$ and $k = \frac{K-K_C}{K_C}$ and keep only the first order. We care about the relation to t , so for small t , the smaller power is more significant, which is in this case the power of a which is $1/4$, giving $\beta = 1/4$:

$$m \sim \sqrt{\frac{-b \pm \sqrt{b^2 - 3ad}}{3d}} \sim \sqrt{\frac{-b_T t \pm \sqrt{b_T^2 t^2 - 3da_T t}}{3d}} \sim t^{1/4} \quad (5)$$

Now let us find α - the critical exponent of the heat capacity. For this, let us remember that the heat capacity can be calculated from the free energy using:

$$C = -\frac{\partial^2 F}{\partial T^2} \quad (6)$$

The free energy is

$$F = V\Phi(m_0, h = 0) = V(am^2 + bm^4 + dm^6) \quad (7)$$

Since $a, b \sim t$ and $m \sim t^{1/4}$, we get:

$$F \sim (t^{3/2} + t^2 + t^{3/2}) \sim t^{3/2} \quad (8)$$

$$C \sim t^{-1/2} \quad (9)$$

So we got $\alpha = 1/2$. Unsurprisingly, we got different critical exponents from the Ising model.

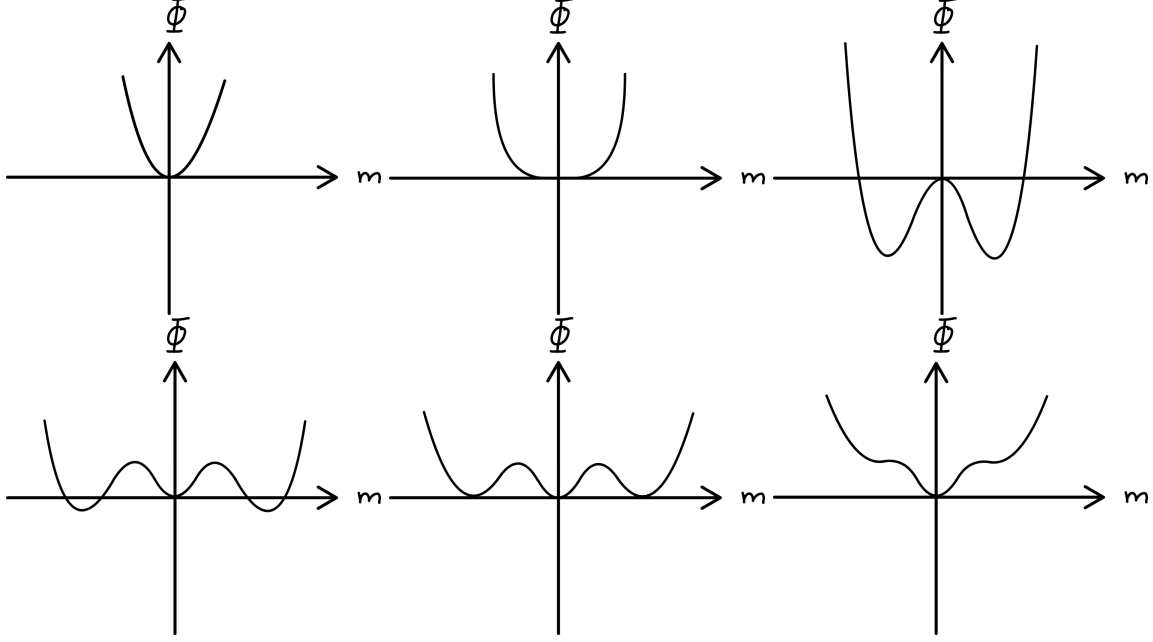


Figure 2: The Landau free energy at different parts of the phase space. Top row: (left) unordered phase with one minimum at $m = 0$ (middle) at the second order phase transition point (right) ordered phase with two minima at $m \neq 0$. Bottom row: (left) ordered phase with two minima at $m \neq 0$ (middle) at the first order phase transition point (right) at the line separating the 3-solutions and 5-solutions areas.

2 Landau Theory for the XY model

Let us consider the XY model:

$$H = \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j - \vec{B} \sum i \vec{S}_i \quad (10)$$

where $S_i = (\cos \theta_i, \sin \theta_i)$ for some real θ_i .

2.1 The Landau Free Energy

In this model, we can define an order parameter $\bar{S}$ which is the average spin configuration over an area. For an external field $B = 0$, there is a rotation symmetry $O(2)$, which means only the size of $\bar{S}$ matters, i.e. we get only even terms in $\bar{S}$. When $B \neq 0$, we get an additional $\vec{B} \cdot \bar{S}$ linear term:

$$\Phi(\bar{S}, T, B) = -\vec{B} \cdot \bar{S} + aS^2 + bS^4 \quad (11)$$

2.2 Critical Exponents

Let us now calculate the critical exponents α, β, γ , and δ . At $B = 0$:

$$\Phi(\bar{S}, T, B) = aS^2 + bS^4 \quad (12)$$

This is very similar to the Ising model with S instead of m , and so we expect the exponents to be the same. So, looking at $b > 0$, since that is where the system has minima, we get:

$$S_0^2 = \begin{cases} \pm \frac{a}{2b} & a < 0 \\ 0 & a > 0 \end{cases} \quad (13)$$

Which means a acts like $t = \frac{T-T_c}{T_c}$. On the other hand, b must be greater than zero, and so it has a zeroth order. So, around $t = 0 \rightarrow a = 0$, in leading order:

$$a \approx a_0(T - T_c) \quad b \approx b_0 \quad (14)$$

This gives:

$$S_0 = \sqrt{\frac{2a}{b}} \approx \sqrt{\frac{a_0 T_c t}{2b_0}} \sim t^{1/2} \quad (15)$$

So the first critical exponent is $\beta = 1/2$. Next, let us look at the susceptibility χ at $T > T_c$. For this, we go back to $B \neq 0$, and keeping again the first two terms, we can set $b = 0$. In this case:

$$\bar{S}_0 = \frac{1}{2a} \bar{B} \approx \frac{1}{2a_0 T_c t} \bar{B} \quad (16)$$

And so

$$\chi = \nabla_B \bar{S} = \frac{1}{2a_0 T_c t} \sim t^{-1} \quad (17)$$

Which gives, as expected $\gamma = 1$. Next let us calculate α . The free energy is given by

$$F = V\Phi(S_0, B = 0) = V \left(\pm \frac{2a^2}{b} + \frac{4a^2}{b} \right) \sim V \frac{a^2}{b} \sim t^2 \quad (18)$$

So

$$C \sim \text{const} \quad (19)$$

And we got the expected $\alpha = 0$. Finally, let us look at δ , which encodes the ratio of $\bar{S}$ and $\bar{B}$ at the critical point, rather than above it like γ . For this, let us take the model at $t = 0$ and $B \neq 0$. This means, since $a \sim t = 0$, we get:

$$\Phi \sim -\bar{B} \cdot \bar{S} + b_0 S^4 = -\bar{B} \cdot \bar{S} + b_0 S^2 (\bar{S} \cdot \bar{S}) = (-\bar{B} + b_0 S^2 \bar{S}) \cdot \bar{S} \quad (20)$$

$$\nabla_S \Phi = -\bar{B} + b_0 S^2 \bar{S} = 0 \quad (21)$$

$$\bar{S}_0 = \left(\frac{|\bar{B}|}{b_0} \right)^{1/3} \hat{B} \quad (22)$$

Which gives $\delta = 1/3$.

2.3 The susceptibility prefactor ratio

Let us look at the prefactors of the susceptibility above and below the critical point. We calculated χ_+ in the previous section, let us now calculate χ_- for $t < 0$. This time, since it takes part of the minimization, we cannot ignore b . For this, let us take S_0 to be the minimum $S_0^2 = \pm \frac{2a}{m}$ and expand around it $\bar{S} = \bar{S}_0 + \delta \bar{S}$ where $\delta \bar{S}$ encodes the effects of $\bar{B}$. This gives

$$\Phi = -\bar{B} \cdot \delta \bar{S} + a \delta S^2 + 2a \bar{S}_0 \cdot \delta \bar{S} + b \delta S^4 + 4b \delta S^2 \delta \bar{S} \cdot \bar{S}_0 + 2b \delta S^2 S_0^2 + 4b (\bar{S}_0 \cdot \delta \bar{S})^2 + 4b S_0^2 \bar{S}_0 \cdot \delta \bar{S} + f(S_0) \quad (23)$$

Where $f(\bar{S}_0)$ encodes all the information about $\bar{S}_0$ that we do not care about, since we are about to derive by $\delta \bar{S}$. Grouping the terms by order in $\delta \bar{S}$:

$$\Phi = b \delta S^4 + 4b \delta S^2 \delta \bar{S} \cdot \bar{S}_0 + (a + 2b S_0^2) \delta S^2 + 4b (\bar{S}_0 \cdot \delta \bar{S})^2 - \bar{B} \cdot \delta \bar{S} + (2a \bar{S}_0 + 4b S_0^2 \bar{S}_0) \cdot \delta \bar{S} + f(S_0) \quad (24)$$

Since by the definition of $\bar{S}_0$

$$2a \bar{S}_0 + 4b S_0^2 \bar{S}_0 = 2(a + 2b S_0^2) \bar{S}_0 = 0 \rightarrow a + 2b S_0^2 = 0 \quad (25)$$

so the first square term and the second linear term vanish. Keeping only up to second order in $\delta\bar{S}$, we get:

$$\Phi = 4b (\bar{S}_0 \cdot \delta\bar{S})^2 - \vec{B} \cdot \delta\bar{S} + f(S_0) \quad (26)$$

Deriving by $\delta\bar{S}$ we get:

$$\nabla_{\delta S} \Phi = 8b (\bar{S}_0 \cdot \delta\bar{S}) \bar{S}_0 - \vec{B} = 0 \quad (27)$$

Which is solved by:

$$\delta\bar{S} = \frac{1}{8bS_0^2} \vec{B} = \frac{1}{4a_0} \vec{B} \quad (28)$$

And so we get $A_+ = \frac{1}{2a_0}$ and $A_- = \frac{1}{4a_0}$ so the ratio $A_+/A_- = 2$ matches the ratio we found for the Ising model.

3 Critical exponents from real space RG

In this section we used the Migdal-Kadanoff real space RG that we saw in class to find the critical exponents. The starting point for this are the recursion relations

$$\begin{aligned} e^{2K'_1} &= \cosh 4K_1 \\ e^{2K'_0} &= 4 \cosh 4K_1 \end{aligned} \quad (29)$$

Which gives an unstable fixed point, and thus a critical point, at $K_C = 0.305$. We notice that this is not really a point since there is a degeneracy for K_0 . This will come into play shortly.

Around this point we can linearize the recursion relation to get a matrix that shows the local flow in the model space. This flow then describes the change of values around this point, and thus the critical exponents. So, using the first order Taylor expansion around K_C :

$$\begin{aligned} \delta K'_1 &= \frac{dK'_1}{dK_1} \delta K_1 = 2 \tanh(K_C) \delta K_1 \\ k'_0 &\approx 8 \tanh(K_C) \delta K_1 \end{aligned} \quad (30)$$

And the linear version of the system is:

$$\begin{pmatrix} \delta K'_1 \\ \delta K'_0 \end{pmatrix} = \begin{pmatrix} 2 \tanh(K_C) & 0 \\ 8 \tanh(K_C) & 0 \end{pmatrix} \begin{pmatrix} \delta K_1 \\ \delta K_0 \end{pmatrix} \quad (31)$$

The eigenvalues of this transformation are $\lambda = 0, 2 \tanh(K_C)$, of which of course only $\lambda = 2 \tanh(K_C)$ is of interest to us (The $\lambda = 0$ is a result of the degeneracy we talked about before). Its eigenvector is:

$$v = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (32)$$

Which leads us away from the critical point, as expected. We now remember that upon decimation:

$$t' = \lambda t \quad (33)$$

From which we can derive all of the critical exponents that are independent of the magnetic field. For a start the correlation length changes by a factor of $\sqrt{2}$:

$$\xi(t) = \sqrt{2} \xi(\lambda t) \quad (34)$$

So, using the definition of $\xi \propto t^{-\nu}$, we get:

$$1 = \sqrt{2} \lambda^{-\nu} \quad (35)$$

$$\nu = \frac{\ln 2}{2 \ln \lambda} \approx 0.67 \quad (36)$$

The other critical exponent we can calculate is α , since it relates to the energy directly. Since it is the second derivative of the energy with respect to the temperature, it scales as the energy scales but with inverse square to the temperature, giving

$$C(t) \propto \lambda^2 \left(\sqrt{2} \right)^{-d} C(\lambda t) \quad (37)$$

Using $C(t) = t^{-\alpha}$:

$$\lambda^{-\alpha+2} = 2^{-d/2} \quad (38)$$

$$\alpha = 2 - \frac{d \ln 2}{2 \ln \lambda} = 2 - d\nu \approx 0.66 \quad (39)$$

The result we got are not exact. The real values are $\nu = 1$ and $\alpha = 0$, same as calculated using the mean field approximation.

4 The Potts model

In this section we consider the Potts model

$$H = \frac{-J}{N} \sum_{i < j} \delta(\sigma_i, \sigma_j) \quad \delta(x, y) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases} \quad (40)$$

4.1 The Potts model as a MF model

Since the Hamiltonian is symmetric to $i \leftrightarrow j$, we can add to it the same sum just with $j < i$ instead of $i < j$ and divide by 2. Remembering that $\delta(\sigma_i, \sigma_i) = 1$, adding the $i = j$ cases will just add a constant to the Hamiltonian. So, up to a constant, the Potts model can be written as:

$$H = \frac{-J}{2N} \sum_{i,j} \delta(\sigma_i, \sigma_j) \quad (41)$$

Defining a q dimensional vector space such that each spin is a basis vector in that space:

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix} \quad |2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \end{pmatrix} \quad \dots \quad |q\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} \quad (42)$$

We can identify the operator $\delta(\sigma_i, \sigma_j) = \langle \sigma_i | \sigma_j \rangle$. Thus we can write:

$$H = \frac{-J}{2N} \sum_i \langle \sigma_i | \sum_j | \sigma_j \rangle \equiv -\frac{J}{2} \sum_i \langle \sigma_i | \bar{\sigma} \rangle \quad |\bar{\sigma}\rangle = \frac{1}{N} \sum_j | \sigma_j \rangle \quad (43)$$

When we defined $|\bar{\sigma}\rangle$ as the mean magnetization vector. This means that each spin interacts with an average over the entire space, which qualifies this model as a mean field model.

We can thus see why the division by N is required, since without it the energy would not be extensive. For example, if $\sigma_i = 1$ for every spin, we get, from the double sum (in the original Hamiltonian):

$$H = \frac{-J}{2N} N(N-1) = -\frac{J}{2}(N-1) \quad (44)$$

Without the N in the denominator it would not have been linear with respect to N .

4.2 Flavour magnetization

Let us define the k -flavour magnetization

$$m_k = \frac{1}{N} \sum_i \delta(k, \sigma_i) = \langle k | \bar{\sigma} \rangle, \quad k = 1, 2, \dots, q \quad (45)$$

This is just the number of spins of a certain k divided by N . Therefore, the number of spins with magnetization k is $N_k = Nm_k$. The energy is then:

$$E(\{m_k\}) = -\frac{J}{2} \sum_i \langle \sigma_i | \bar{\sigma} \rangle = -\frac{J}{2} \sum_k N_k \langle k | \bar{\sigma} \rangle = -\frac{J}{2} N \sum_k m_k^2 \quad (46)$$

Then, the multiplicity of every configuration $\{m_k\}$ is:

$$\begin{aligned} \mu(\{m_k\}) &= \binom{N}{N_1} \binom{N-N_1}{N_2} \cdots \binom{N-N_1-\cdots-N_{q-2}}{N_{q-1}} = \\ &= \frac{N!}{(N-N_1)!N_1!} \frac{(N-N_1)!}{(N-N_1-N_2)!N_2!} \cdots \frac{(N-N_1-\cdots-N_{q-2})!}{(N-N_1-N_2-\cdots-N_{q-1})!N_{q-1}!} = \\ &= \frac{N!}{N_1!N_2! \cdots N_q!} \end{aligned} \quad (47)$$

Using Stirling's approximation, we can find the entropy $S(\{m_k\}) = \ln \mu(\{m_k\})$:

$$\begin{aligned} S(\{m_k\}) &= N(\ln N - 1) - \sum_k N_k(\ln N_k - 1) = N \ln N - \sum_k N_k \ln N m_k = \\ &= N \ln N - \left(\sum_k N_k \right) \ln N - \sum_k N_k \ln m_k = \\ &= - \sum_k N_k \ln m_k = -N \sum_k m_k \ln m_k \end{aligned} \quad (48)$$

So, the total free energy is

$$F(\{m_k\}) = -N \left(\frac{J}{2} \sum_k m_k^2 - T \sum_k m_k \ln m_k \right) \quad (49)$$

4.3 First order phase transition

Let us now look for first order phase transitions. For this, let us assume that all k -flavour magnetizations are the same except for one which is slightly different:

$$m_k = \begin{cases} \bar{m} - \frac{\Delta m}{q-1} & k \neq d \\ \bar{m} + \Delta m & k = d \end{cases} \quad (50)$$

Where $\Delta m \ll \bar{m}$. The free energy is then:

$$\begin{aligned} \frac{1}{N} F(\{m_k\}) &= (q-1) \left(-\frac{J}{2} \left(\bar{m} - \frac{\Delta m}{q-1} \right)^2 + T \left(\bar{m} - \frac{\Delta m}{q-1} \right) \ln \left(\bar{m} - \frac{\Delta m}{q-1} \right) \right) + \\ &+ \left(-\frac{J}{2} (\bar{m} + \Delta m)^2 + T (\bar{m} + \Delta m) \ln (\bar{m} + \Delta m) \right) \end{aligned} \quad (51)$$

For a first order phase transition we look for when the prefactor of Δm^3 is non-zero. Expanding the $\ln$ expressions:

$$\ln\left(\bar{m} - \frac{\Delta m}{q-1}\right) \approx \ln \bar{m} - \frac{\Delta m}{\bar{m}(q-1)} - \frac{1}{2} \frac{\Delta m^2}{\bar{m}^2(q-1)^2} - \frac{1}{3} \frac{\Delta m^3}{\bar{m}^3(q-1)^3} + O(\Delta m^4) \quad (52)$$

$$\ln(\bar{m} + \Delta m) = \ln \bar{m} + \ln\left(1 + \frac{\Delta m}{\bar{m}}\right) \approx \ln \bar{m} + \frac{\Delta m}{\bar{m}} - \frac{1}{2} \frac{\Delta m^2}{\bar{m}^2} + \frac{1}{3} \frac{\Delta m^3}{\bar{m}^3} + O(\Delta m^4) \quad (53)$$

We get the Δm^3 term:

$$c\Delta m^3 = T \left(-\bar{m} \frac{1}{3} \frac{\Delta m^3}{\bar{m}^3(q-1)^3} + \frac{\Delta m}{q-1} \frac{1}{2} \frac{\Delta m^2}{\bar{m}^2(q-1)^2} + \bar{m} \frac{1}{3} \frac{\Delta m^3}{\bar{m}^3} - \Delta m \frac{1}{2} \frac{\Delta m^2}{\bar{m}^2} \right) \quad (54)$$

And so the prefactor is:

$$c = \frac{5T}{6\bar{m}^2} \left(1 - \frac{1}{(q-1)^3} \right) \neq 0 \rightarrow q > 2 \quad (55)$$

So the for $q = 2$ there is only a second order phase transition, while for $q > 2$ there is also a first order phase transition.

4.4 Landau theory with a complex order parameter

Conversly, we can describe the Potts model using a single complex order parameter m . To understand how to do this, let us notice that the system is invariant to permutation of the basis vectors $|k\rangle$. In other words, for every i, j , switching all spins $|i\rangle$ to state $|k\rangle$ and vice versa, the energy will not change. Part of this symmetry is the symmetry of rotation - moving every index one position up or down. This rotational symmetry gives us the option to describe each magnetization as a complex q -root of 1. Then, a complex order parameter m will be the linear combination of those roots:

$$m = \sum_k m_k e^{i2\pi k/q} \quad (56)$$

And it obeys the symmetry

$$m \rightarrow e^{i\theta} m \quad \theta = \frac{2\pi k}{q} \quad k = 0, \dots, q-1 \quad (57)$$

The Landau free energy then includes phase invariant terms in the polynomial, while also including terms that have powers multiples of q . For small m , taking the lowest order, we get:

$$\Phi(m) = |m|^2 + |m|^4 + m^q + m^{*q} \quad (58)$$

This will give first order transition only for $q = 3$. This is different from what we recieved in the previous section, because this model reduces the index permutation symmetry to an index rotation symmetry.

4.5 Monte-Carlo simulation

To cap off this section, we preform a Monte Carlo simulation to view the actual results of the mean field Potts model. We simulated a 1d model with $N = 180$ periodic boundary conditions. The selected order parameter was σ , the standard diviation of the different m_k . This was selected as in the ordered phase it is maximal while in the disordered phase it is zero, and it related to ΔM that was saw here was a valid order parameter, but is also simpler to calcualte.

The simulations used Single Spin Metropolis dynamics, where instead of flipping the spin the dynamics changes it to a random spin of a different value. We ran 5 parallel simulations for 500 iterations after convergence. Convergence was defined by a threshold of the diviation between the five states and their

average change during the last ten iterations. The simulation was ran repeatedly with starting from $T = 0.02$ in jumps of $\Delta T = 0.02$ up to $T = 2.0$. For each simulation, the different m_k were recorded, then the total states after convergence were plotted in a histogram and fit to a gaussian. The mean and standard deviation of the gaussian were taken to be the order parameter and its measurement error. The final graphs of σ/T can be found in figure 3.

It can be seen that for $q = 2$ there is a clean second order phase transition, while for $q = 3$ there is a first order phase transition, but it is not as clear. While there is a visible critical and a jump in the order parameter, there are several jumps before. These are results of the metastable states in the finite system. Since the $q = 3$ has a metastable state at $\sigma = 0$, and because the system has a finite number of spins, it might, in lower temperature, be stuck in that state. This results in the three anomalies that can be seen in the graph. All in all the results of the measurements agree with our predictions.

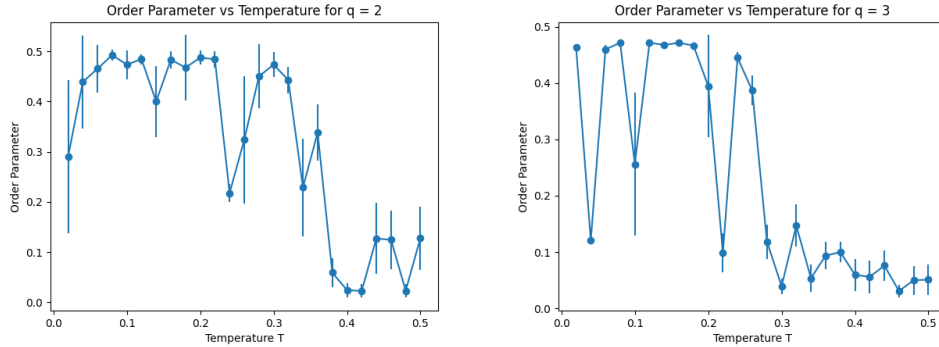


Figure 3: The order parameter σ as a function of temperature for the Potts model with $q = 2$ (left) and $q = 3$ (right).